# E-protein regulatory network links TCR signaling to effector Treg cell differentiation

Xiaojuan Han[a,b], Huarong Huang[a,b], Ping Gao[a,b], Qi Zhang[a], Xinyuan Liu[a,b], Baoqian Jia[a], Warren Strober[c], Baidong Hou[d], Xuyu Zhou[a,e], George Fu Gao[a,e,f,g], and Fuping Zhang[a,e,1]

[a]Chinese Academy of Sciences (CAS) Key Laboratory of Pathogenic Microbiology and Immunology, Institute of Microbiology, Chinese Academy of Sciences, 100101 Beijing, China; [b]College of Life Science, University of Chinese Academy of Sciences, 100049 Beijing, China; [c]Mucosal Immunity Section, Laboratory of Host Defenses, National Institute of Allergy and Infectious Diseases, NIH, Bethesda, MD 20892; [d]CAS Key Laboratory of Pathogenic Microbiology and Immunology, Institute of Biophysics, Chinese Academy of Sciences, 100101 Beijing, China; [e]Savaid Medical School, University of Chinese Academy of Sciences, 101408 Beijing, China; [f]National Institute for Viral Disease Control and Prevention, Chinese Center for Disease Control and Prevention, 102206 Beijing, China; and [g]Research Network of Immunity and Health, Beijing Institutes of Life Science, Chinese Academy of Sciences, 100101 Beijing, China

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**T cell antigen receptor (TCR) signaling is essential for the differentiation and maintenance of effector regulatory T (Treg) cells. However, the contribution of individual TCR-dependent genes in Treg cells to the maintenance of immunotolerance remains largely unknown. Here we demonstrate that Treg cells lacking E protein undergo further differentiation into effector cells that exhibit high expression of effector Treg signature genes, including IRF4, ICOS, CD103, KLRG-1, and RORγt. E protein-deficient Treg cells displayed increased stability and enhanced suppressive capacity. Transcriptome and ChIP-seq analyses revealed that E protein directly regulates a large proportion of the genes that are specific to effector Treg cell activation, and importantly, most of the up-regulated genes in E protein-deficient Treg cells are also TCR dependent; this indicates that E proteins comprise a critical gene regulatory network that links TCR signaling to the control of effector Treg cell differentiation and function.**

E protein | effector Treg cell | TCR signaling | gene regulatory network | differentiation and function

It is well established that regulatory T cells (Treg cells) play a pivotal role in maintaining immune tolerance and thus in preventing autoimmunity and chronic inflammation (1). These cells can be subdivided into different subsets with somewhat unique characteristics. One type of subset classification is based on site of development and consists of a natural Treg subset that develops in the thymus (called nTreg or tTreg cells) and a peripheral Treg cell subset that develops in the periphery (called iTreg or pTreg cells) (2). Another type of subset classification is based on function. This classification recognizes a "central" Treg population that is equivalent to naive $CD4^+$ T cells with respect to various markers and circulatory patterns and that differs from one or more "effector" Treg cell populations that exhibit enhanced regulatory function and migration through nonlymphoid tissues (3–5). Such effector Treg cells make up a minor fraction of Treg cells in the circulation and secondary lymphoid organs and are referred to as "activated" Treg cells in some studies because they share phenotypic features with activated conventional T cells. As such, they are variously defined as $CD62L^{low}CCR7^{low}CD44^{hi}KLRG1^+$ cells, $ICOS^+IRF4^+CD103^+$ cells, or $CD45RA^{low}CD25^{hi}$ cells, depending on the study.

Importantly, Treg cells comprising the effector subpopulation defined above are thought to have encountered antigens and have undergone T cell antigen receptor (TCR) stimulation more recently than central Treg cells (6–9). This coincides with the fact that TCR signaling in differentiated Treg cells is essential to Treg cell homeostasis, suppressor function, and signature gene expression, especially to those signature genes that produce Treg cell effector molecules (10–12). It should be noted, however, that the underlying molecular mechanisms downstream of TCR signaling that account for the above Treg cell characteristics are still poorly understood. In addition, the contributions of individual TCR-dependent genes in Treg cells to maintain immunotolerance in the steady state and that restrain immune responses directed against commensal bacteria, environmental antigens, and pathogens, also require further elucidation.

One important way in which effector Treg cell differentiation may be regulated is via the activity of E proteins, a family of transcriptional activators or repressors that bind to E-box sites at transcriptional sites and thus determine gene expression (13). Previous studies have shown that down-regulation of E-protein activity by TCR signaling accompanies the development of $Foxp3^+$ regulatory T cells (14, 15), whereas increased E-protein activity accompanies Treg cell dysfunction (16). These observations suggested that recognition of self-antigen in matured Treg cells may further down-regulate E-protein activity, thus driving Treg cells to differentiate toward particular specificities that may potentiate their suppressive capacity during immunological challenges. Nevertheless, whether E protein regulates matured Treg cell activation or whether E protein directly regulates certain activated Treg signature gene expression has never been demonstrated.

To directly address the requirement of E-protein activity in the regulation of Treg cell homeostasis, we studied Treg cells in $E2A^{fl/fl}HEB^{fl/fl}$*Foxp3*-Cre-GFP mice with E-protein deletion specific to Treg cells. Notably, we found that deficiency of E protein in differentiated Treg cells results in up-regulated expression of

## Significance

**Effector Treg cells comprise the subset of the Treg cell population that exhibits enhanced regulatory function. Whereas the induction and maintenance of this subset are known to depend on TCR signaling, the underlying molecular mechanisms downstream of such signaling and the contributions of individual TCR-dependent genes to effector Treg cell generation are still poorly understood. In the studies described here differentiated Treg cells in which E-protein (E2A/HEB) expression has been deleted were utilized to demonstrate that E proteins are transcriptional suppressors of a large number of genes associated with effector Treg cell differentiation, localization, function, and proliferation. Thus, this finding indicates that continuous TCR signals modulating E-protein activity is a major mechanism underlying Treg cell acquisition of effector functions.**

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[1]To whom correspondence should be addressed. Email: zhangfp@im.ac.cn.

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various effector Treg cell markers such as CD103, ICOS, IRF4, KLRG1, RORγt, as well as enhanced Treg cell stability and suppressive capacity. Further study demonstrated that E-protein activity was essential for Treg cells to maintain the expression of a large proportion of genes found to be expressed almost exclusively in effector Treg cells. In addition, gene expression profile analysis indicated that a substantial amount of effector Treg signature genes regulated by E protein are also regulated by TCR. E proteins regulate effector Treg cells by directly binding to a large number of genes that associated with effector Treg cell differentiation, localization, function, and proliferation. Thus, our findings demonstrated that downstream of TCR signaling, E-protein activity plays an essential role in the regulation of Treg cell activation.

## Results

**Treg Cell Homeostasis Is Regulated by Specific Deletion of E Proteins in Mature Treg Cells.** To explore the specific role of E proteins in mature Treg cells, we crossed *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$ mice with mice that have a bacterial artificial chromosome transgene encoding green fluorescent protein (GFP) and humanized Cre recombinase (hCre) under the control of the Foxp3 promoter (Foxp3-GFP-hCre mice) (17, 18). Thus, the resultant *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*-GFP-hCre mice (designated hereafter as *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$ mice) exhibited specific deletion of E2A and HEB in Treg cells but not in CD4$^+$ conventional T cells (*SI Appendix*, Fig. S1*A*). We did not observe changes in development of CD4$^+$ and CD8$^+$ T cells or the proportion of Treg cells in the thymus (*SI Appendix*, Fig. S1 *B* and *C*). However, we did observe a small, but statistically significant, increase in the proportion of Treg cells in the spleen, peripheral lymph nodes (pLN), and mesenteric lymph nodes (mLN) (*SI Appendix*, Fig. S1*D*), demonstrating a modest, but integral role of E protein in the regulation of peripheral Treg cell homeostasis. The lack of increase in Foxp3$^+$ cells in the thymus of *E2A*$^{f/f}$*HEB*$^{f/f}$ mice was likely due to the fact that Foxp3$^+$ cells in the thymus are newly developed Tregs in which a TCR-driven decline in E-protein levels determines the number of Tregs rather than Foxp3(Cre)-dependent E-protein deletion. To investigate this possibility, we took advantage of *Rosa26*-loxP-Stop-loxP-YFP (*R26*$^{YFP}$) mice, in which YFP is expressed only after Cre is expressed. Accordingly, we crossed *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$ *Foxp3*$^{Cre}$ with *R26*$^{YFP}$ mice, in which Foxp3-expressing cells that are E-protein gene deleted are labeled with YFP, so that YFP$^+$ cells are E-protein knockout (KO) Tregs. As expected, we found that most of the Tregs in thymus of *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ and WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ mice were YFP-negative (*SI Appendix*, Fig. S1*E*), indicating that at this site TCR-driven decreases in E protein are largely responsible for Treg development rather than Foxp3 (Cre)-dependent E-protein gene deletion. In contrast, most of the Treg cells in the periphery are YFP-positive, indicating that the cells have undergone E-protein deletion as a result of Foxp3 (Cre) expression. Thus, the lack of increase of Tregs in the thymus of *E2A*$^{f/f}$*HEB*$^{f/f}$ mice is probably due to the fact that the Tregs in thymus are subject to a lesser degree of Foxp3-Cre mediated E-protein deletion than cells in the periphery.

In related studies we found that the GI lamina propria, a mucosal lymphoid site that is exposed to an abundance of foreign antigens and that requires Treg cells for maintenance of intestine homeostasis, exhibited a dramatic increase in Treg cell frequency in *E2A*$^{f/f}$*HEB*$^{f/f}$ mice and this increase was substantially greater than the increase in spleen and lymph nodes (Fig. 1*A*). Similarly, we found that Treg cell increases were also greater in liver and lung than in lymphoid organs (Fig. 1*A*). Recognizing that Treg cell development in the lamina propria is highly TGF-β dependent, we investigated the role of E protein on iTreg cell differentiation in vitro and found that upon stimulation with TCR and TGF-β, cells lacking E protein gave rise to an increased proportion of Foxp3$^+$ cells (*SI Appendix*, Fig. S1*F*). Finally, we observed an increase in the percentages of both Helios$^+$Foxp3$^+$ cells and Helios$^-$Foxp3$^+$ cells in lamina propria of *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$ mice compared with that of WT mice (*SI Appendix*, Fig. S1*G*). Since Helios is a marker of nTreg cells (19), this suggests that E protein regulates both nTreg cells as well as iTreg cells.

Collectively, these results indicate that the E-protein level in mature Treg cells plays a substantial role in regulating Treg cell homeostasis particularly at nonlymphoid organs exposed to TCR signaling by ambient antigens.

**E Proteins Regulate Treg Cell Homeostasis in a Cell-Intrinsic Manner.** To determine if the effects of E protein on Treg cell homeostasis are cell-intrinsic, we generated mixed bone marrow (BM) chimeric mice by mixing equal numbers of BM cells from CD45.1$^+$ CD45.2$^+$ WT *Foxp3*$^{Cre}$ mice and CD45.1$^+$ *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$ *Foxp3*$^{Cre}$ mice and then transferred the cell mixtures into lethally irradiated CD45.2$^+$ mice. Analyses of the number of Treg cells from WT (CD45.1$^+$CD45.2$^+$) or E protein-deficient (CD45.1$^+$) cells in lymphoid organs 2 mo after cell transfer showed that E protein-deficient Treg cells were significantly increased compared with WT Treg cells (Foxp3$^+$CD45.1$^+$/Foxp3$^+$CD45.1$^+$ CD45.2$^+$ ratio: 1.5) (Fig. 1*B*). In addition, in nonlymphoid organs of these mice, an even more pronounced increase in E protein-deficient Treg cells was observed (Foxp3$^+$CD45.1$^+$/Foxp3$^+$CD45.1$^+$CD45.2$^+$ ratio: 2.5) (Fig. 1*B*). To investigate E-protein effects on Treg cell homeostasis under inflammatory conditions, we immunized mixed bone marrow chimeric mice with myelin oligodendrocyte glycoprotein (MOG) to induce experimental allergic encephalomyelitis (EAE) or administered dextran sodium sulfate (DSS) in drinking water to induce colitis. As expected,

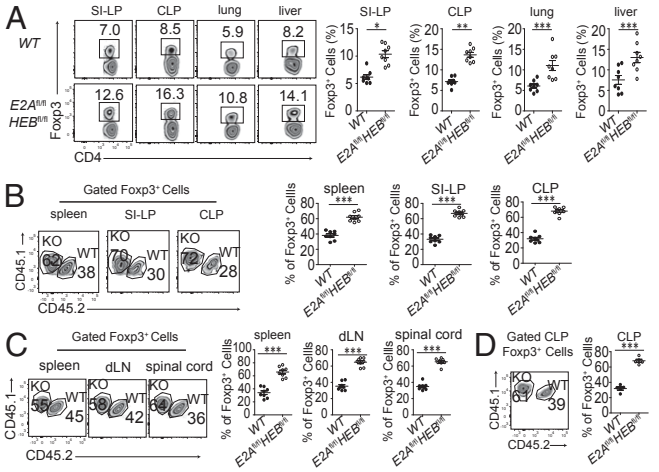


**Fig. 1.** E protein regulates Treg cell homeostasis in a cell-intrinsic manner. (*A*) Representative flow cytometric analysis (*Left*) and the percentages (*Right*) of Foxp3$^+$ cells within CD4$^+$ cells in indicated organs of WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice. (*B–D*) Mixed BM cells from WT*Foxp3*$^{Cre}$ mice (CD45.1$^+$CD45.2$^+$ cells) and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice (CD45.1$^+$ cells) were transferred into lethally irradiated recipient (CD45.2$^+$) mice and were analyzed 2 mo after reconstitution. (*B*) Data on the *Left* show representative flow cytometry analysis of WT and KO Treg cells in chimeric mice after gating on Foxp3$^+$ cells and data on the *Right* show frequencies of WT (CD45.1$^+$CD45.2$^+$) and KO (CD45.1$^+$) cells among the reconstituted Tregs (CD4$^+$Foxp3$^+$) in indicated organs. (*C*) Two months after reconstitution, EAE was induced in the chimeric mice; 24 d after induction, Foxp3$^+$ cells were analyzed by flow cytometry; data on the *Left* show representative flow cytometry analysis of WT and KO Treg cells in chimeric mice after gating on Foxp3$^+$ cells and data on the *Right* show the frequencies of WT (CD45.1$^+$CD45.2$^+$) and KO (CD45.1$^+$) cells among the reconstituted Tregs (CD4$^+$Foxp3$^+$) in indicated organs. (*D*) DSS-colitis was induced in the chimeric mice; 10 d after induction Foxp3$^+$ cells were analyzed by flow cytometry; representative flow cytometry analysis (*Left*) and frequencies (*Right*) of WT (CD45.1$^+$CD45.2$^+$) and KO (CD45.1$^+$) cells in reconstituted Tregs (CD4$^+$Foxp3$^+$) in the CLP. Data in *A–D*, *Left* are representative of at least three independent experiments. Data in *A–D*, *Right* are pooled from at least three independent experiments. Each symbol represents data from one mouse. Graph shows mean ± SD (*$P$ < 0.05, **$P$ < 0.01, ***$P$ < 0.001).



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24 d after induction, we observed a markedly increased number of E protein-deficient Treg cells vs. WT Treg cells infiltrating the spinal cord as well as in spleen and draining lymph nodes (dLN) (Fig. 1*C*) of mice with EAE. A similar observation was obtained in colon lamina propria (CLP) of chimeric mice 10 d after DSS administration (Fig. 1*D*). It should be noted that the frequency of Treg cells in KO mice without inflammation was comparable to that of KO mice with EAE or colitis (compare Fig. 1*B* with Fig. 1 *C* and *D*). These studies thus indicated that the increased percentage of Foxp3$^+$ cells among CD4$^+$ cells due to E-protein deletion is maintained in a cell-intrinsic manner under both steady-state and inflammatory conditions.

**E-Protein Deletion Stabilizes Foxp3 Expression in Matured Treg Cells.** While Treg cells are generally stable, under lymphopenic or inflammatory conditions a fraction of these cells lose Foxp3 expression and undergo other phenotypic changes, such as acquisition of ability to secrete cytokines that mediate diverse effector functions (18). To elucidate the effect of E proteins on Treg cell stability, we used cells from *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice and WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ mice as described above to compare the effect of E-protein deletion on the generation of YFP$^+$Foxp3$^-$ (GFP$^-$) "exTreg" cells. In initial studies, we compared the impact of inflammatory cytokines on the stability of highly purified E protein-deficient vs. WT spleen YFP$^+$ cells after culture of the cells in the presence of IL-2 plus various proinflammatory cytokines (IL-12, IFN-γ, IL-4, or IL-6). We found that even in the presence of optimal amounts of IL-2, the stability of WT Treg cells was severely compromised compared with that of E protein-deficient Treg cells as indicated by the greater percentage of exTreg cells that appeared in the WT cells (Fig. 2*A*). Intriguingly, the exTreg cells from the WT mice produced more IFN-γ (*SI Appendix*, Fig. S2*A*) and IL-17A (*SI Appendix*, Fig. S2*B*) than exTreg cells from E protein-deficient mice; this suggests that E proteins may promote exTreg cells to become pathogenic T helper cells during inflammations such as EAE. In addition, we observed a lower percentage of YFP$^+$Foxp3$^-$ (exTreg) cells among YFP$^+$ cells in the small intestine lamina propria and CLP of *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice than in WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ mice (Fig. 2*B* and *SI Appendix*, Fig. S2*E*). However, in this case the difference, while significant, was less than that observed in cultured cells. In contrast, there were no differences in the percentage of exTregs in the spleen and pLN cells of *E2A*$^{f/f}$*HEB*$^{f/f}$*R26*$^{YFP}$ mice and WT mice perhaps because cells in these tissues are less subject to antigen stimulation than those in the mucosal tissues (*SI Appendix*, Fig. S2*F*).

To further investigate the role of E protein on the stability of Treg cells, we transferred highly purified YFP$^+$ Treg cells from spleen and lymph nodes of *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ or WT*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice into *Rag2*$^{-/-}$ recipients and assessed Foxp3$^+$ (GFP$^+$) cell stability under lymphopenic conditions 4 wk after cell transfer. Consistent with the findings above, fewer E protein-deficient Treg cells lose Foxp3 expression than WT Treg cells, indicating that the E protein- deficient Treg cells are more stable under lymphopenic conditions (Fig. 2*C*).

To confirm the above findings concerning Treg cell stability, we determined the role of E-protein activity in maintaining Treg cell stability in cells from *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice, i.e., mice in which E-protein effects are increased because they are not blocked by Id protein. Recognizing that mice with deletion of Id2/Id3 in matured Treg cells exhibit spontaneous inflammation that might have a secondary effect on Treg cell stability (16), we examined the effect of Id proteins on Treg cell stability under the noninflammatory conditions in mixed bone marrow chimeric mice. Bone marrow cells from WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ and *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice were mixed at 1:1 ratio and transferred into lethally irradiated recipient mice; subsequently (i.e., at 8 wk after cell transfer when the recipient mice were robust and thus apparently free of inflammation), we analyzed the fraction of exTreg cells. We found that the percentage of exTreg

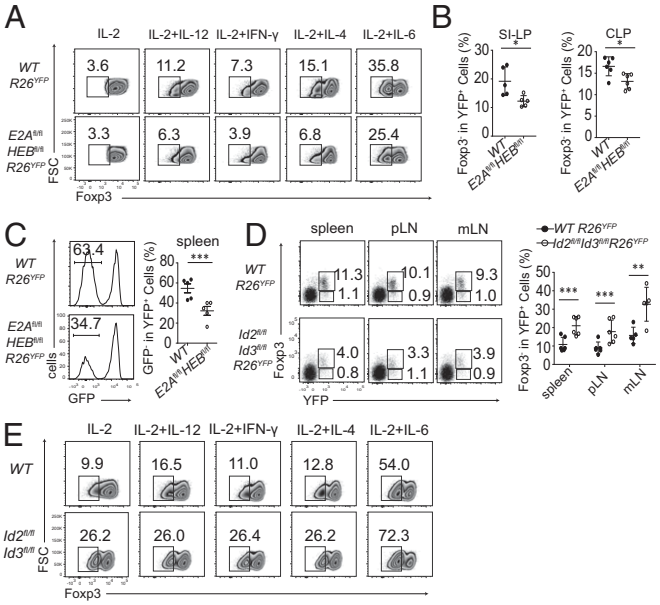


**Fig. 2.** E-protein deletion stabilizes Foxp3 expression in matured Treg cells. (*A*) Highly purified YFP$^+$ Treg cells from WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice were activated and cultured in the presence of IL-2 plus indicated cytokine for 4 d; Foxp3 expression (among YFP$^+$ cells) was then analyzed by flow cytometry. (*B*) The frequency of Foxp3$^-$ in YFP$^+$ cells (*Right*) in indicated organs of WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice. (*C*) Highly purified Treg (YFP$^+$) cells from WT *Foxp3*$^{Cre}$*R26*$^{YFP}$*Foxp3*GFP or *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$*Foxp3*GFP mice were sorted by FACS and were transferred into *Rag2*$^{-/-}$-recipient mice. GFP expression among YFP$^+$ cells was analyzed by flow cytometry 4 wk after cell transfer (*Left*) and the frequency of GFP$^-$ (Foxp3$^-$) among YFP$^+$ cells is shown (*Right*). (*D*) Mixed BM chimeric mice were generated by transferring BM cells from WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ and *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice into lethally irradiated recipient mice. Flow cytometry analysis of Foxp3 and YFP expression in CD4$^+$ cells (*Left*) and the frequency of Foxp3$^-$ cells among the YFP$^+$ cells (*Right*) of WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ and *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ donors in indicated organs in the chimeric mice are shown. (*E*) Highly purified CD25$^{hi}$ CD4$^+$ Treg cells from WT*Foxp3*$^{Cre}$ and *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ mice were activated and cultured in the presence of IL-2 plus indicated cytokine for 4 d; Foxp3 expression was analyzed by flow cytometry. (*A*, *E*, and *B–D*, *Left*) Data are representative of at least three independent experiments. (*B–D*, *Right*) Data are pooled from at least three independent experiments. Each symbol represents data from one mouse. Graph shows mean ± SD (*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$).

(Foxp3$^-$) cells among YFP$^+$ cells was greatly increased in spleen, pLN, and mLN cells derived from Id protein-deficient mice compared with cells derived from WT mice (Fig. 2*D*). In companion studies we cultured highly purified CD25$^+$CD4$^+$ Treg cells from WT*Foxp3*$^{Cre}$ and *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ mice in the presence of IL-2 and proinflammatory cytokines as described above, and found that the inflammatory cytokines compromised Foxp3 expression in Id protein-deficient Treg cells to a greater extent than in WT Tregs (Fig. 2*E*); in addition, these exTreg cells from *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ mice produced increased amounts of cytokines (IFN-γ, IL-4) compared with that of WT exTreg cells (*SI Appendix*, Fig. S2 *C* and *D*). Collectively, these studies provide strong evidence that lack of E-protein expression in *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice and increased E-protein activity in *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ mice affects Treg stability in reciprocal ways, most likely via underlying effects on survival or proliferation of Foxp3$^+$ cells.

**E Protein-Deficient Treg Cells Exhibit Enhanced Immunosuppressive Function both in Vitro and in Vivo.** Previous studies have demonstrated that Treg cells have a prominent capacity to limit inflammation during experimental autoimmune diseases such as EAE or cell-transfer colitis. Based on the above findings, we

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reasoned that E-protein deletion specific to Treg cells would lead to enhanced Treg cell function. To investigate this possibility, we induced EAE in WT and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice and found that *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice were more resistant to EAE, as evidenced by lower disease scores, not only during the initial progressive phase of disease, but also during the remitting-relapsing phase of disease occurring after peak disease (Fig. 3*A*). This finding could be attributed to differences in Treg cell migration since significantly more Foxp3$^+$ cells were present in the spinal cords of *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice than those of WT *Foxp3*$^{Cre}$ mice (Fig. 3*B*). Of interest, there were more MOG-specific Treg cells in the *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice than that of WT *Foxp3*$^{Cre}$ mice (Fig. 3*C*); accordingly, MOG-specific CD4$^+$ T cells were greatly decreased in *E2A*$^{f/f}$*HEB*$^{f/f}$ mice compared with that of WT mice (Fig. 3*C*), suggesting that E protein-deficient Treg cells are more responsive to antigen stimulation, and antigen-specific E protein-deficient Treg cells exhibit enhanced suppressive capacity. *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice spinal cords harbored significantly fewer IL-17– and IFN-γ–producing cells (Fig. 3 *D* and *E*), which also suggests that *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice were more resistant to EAE.

As *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice have an increased percentage of Treg cells in the spinal cords, the increased suppressive function of Treg cells in *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice with EAE may simply be due to the presence of an increased Treg cell pool rather than because the Treg cells are more suppressive. To clarify this issue, we performed in vitro suppressive assays in which titrated Treg cells from WT or *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ were cocultured with an equal number of carboxyfluorescein diacetate succinimidyl ester (CFSE)-labeled CD4$^+$ T cells and the CD4$^+$ T cell proliferation was assessed. We found that E protein-deficient Treg cells had an increased capacity to suppress conventional CD4$^+$ T cell proliferation compared with WT Treg cells. Thus, Treg cells with E-protein deletion do exhibit increased intrinsic suppressive activity (Fig. 3*F*).

To further investigate the suppressive function of E protein-deficient Treg cells in vivo, we turned to the well-characterized cell transfer-induced colitis model. Here, we compared colitis developing in *Rag2*$^{-/-}$ mice following transfer of naïve CD45RB$^{hi}$ CD4$^+$ T cells together with Treg cells from WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ or *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice. Recipient mice cotransferred E protein- deficient Treg cells exhibited less colitis than mice cotransferred WT Treg cells as indicated by their increased weight gain (Fig. 3*G*), decreased cell infiltration in the colon (*SI Appendix*, Fig. S2*G*), and by the dramatic decrease of both total CD4$^+$ T cells and total IL-17– and IFN-γ–producing CD4$^+$ T cells (Fig. 3 *H*–*J* and *SI Appendix*, Fig. S2*H*). Importantly, although the Treg cell percentages were increased in spleen, mLN, and colon of mice that were transferred E protein-deficient Tregs (Fig. 3 *H* and *K*) due to decreased total CD4$^+$ T cell numbers (Fig. 3*H*) compared with mice transferred WT Treg cells, the absolute numbers of Treg cells in spleen, mLN, and colon were comparable in the two groups (Fig. 3*K*). Thus, these data further support the view that decreased E-protein levels in Treg cells result in Treg cells with increased regulatory function.

**E Protein Globally Regulates Effector Treg Cell Signature Gene Expression.** The above findings showing that E protein-deficient Treg cells exhibit increased homeostasis, enhanced suppressive function and increased stability prompted us to hypothesize that E protein might control a distinct, lineage-specific, transcriptional program in Treg cells. To investigate this possibility, we used high-throughput RNA-sequencing (RNA-seq) analysis to compare the global gene-expression patterns of E protein-deficient and -sufficient Treg cells. Recognizing that Treg cells are constantly being stimulated by self-antigens and that E proteins are regulated by TCR stimulation, we compared E protein-deficient and -sufficient Treg cells without stimulation [KO vs. WT (no TCR hereafter)] or stimulated with anti-CD3/CD28 for 48 h [KO vs. WT (TCR 48 h hereafter)] in the studies.

The RNA-seq study showed that 801 genes were up- or down-regulated at least twofold in E protein-deficient Treg cells relative to the genes in WT Treg cells (Fig. 4*A*, *Top* and Fig. 4*B*, Venn A). These genes were termed "E protein-dependent genes," whereas 1,611 genes changed expression at least twofold in E protein-deficient Treg cells subjected to TCR stimulation relative to genes in WT Treg cells also subjected to TCR stimulation (Fig. 4*A*, *Bottom* and Fig. 4*B*, Venn B). Meanwhile, 6,606 gene expressions were changed in WT Treg cells after TCR stimulation (WT vs. WT-48 h), which were termed "TCR-dependent genes" (Fig. 4*B*, Venn C). In further analysis, we found that a similar proportion of genes regulated by E protein [WT vs. KO cells (Fig. 4*B*, Venn A) and WT-TCR 48-h vs. KO-TCR 48-h cells (Fig. 4*B*, Venn B)] were also TCR dependent (Fig. 4*B*, *Right Top*, Venn A overlap C, 433/801 genes = 54% and Fig. 4*B*, *Right Bottom*, Venn B overlap C, 887/1,611 genes = 55%, respectively). This similarity in the proportion of genes regulated by E protein alone and by E protein plus TCR stimulation suggests that the effects of E protein on gene regulation are parallel to or slightly amplified by TCR stimulation. In addition, comparison of E protein and TCR-dependent genes showed that most of the genes up-regulated in E protein-deficient Treg cells were also TCR-dependent genes (Fig. 4*C*). This suggests that a large proportion of Treg cell-specific gene expression downstream of TCR signaling, is dependent on E protein and therefore, the E-protein regulatory network ties TCR signaling to matured Treg cell homeostasis.

Examination of the specific genes up-regulated in KO vs. WT (TCR 48-h) disclosed several genes encoding transcription factors shown previously to be involved in the differentiation and function of effector Treg cells such as *Nfat5*, *Ikzf3*, and *Irf4*, as well as genes encoding factors with a similar potential function, *Itgae*, *Icos*, *Cd44*, *and Lag3* (Fig. 4*D*). In addition, genes encoding secreted factors associated with Treg cell suppressive function, including *Granzyme B*, *Il10*, and *Fgl2* were also significantly increased in KO vs. WT (TCR 48-h) as were genes involved in the TGF-β signaling pathways, such as *Tgfbr3* and *Smad3* (Fig. 4*D*). In a related vein, the expression of RORγt was increased in E protein-deficient Treg cells following TCR stimulation. This is significant because an effector Treg cell lineage subset that expresses both Foxp3$^+$ and RORγt$^+$ has been shown to have enhanced suppressive capacity and increased ability to regulate Th17-mediated immune responses (20–22). Finally, several chemokine and chemokine receptor or integrin-encoding genes were up-regulated in the E protein-deficient cells subjected to TCR stimulation, suggesting that such Treg cells have an increased ability to migrate to inflammatory sites (Fig. 4*D*).

The up-regulation of specific genes in E protein-deficient cells subjected to TCR stimulation noted above was accompanied by the up-regulation of a large number of genes involved in more general cell functions. This includes genes encoding ribosome RNA-processing genes, indicating that protein translation was greatly up-regulated by E-protein deletion in these cells; in addition, consistent with gene ontology (GO) analysis, genes correlating with cell proliferation, cell cycling, and cell apoptosis, such as *cdk6*, *cdkn1a*, and *Bcl2* (Fig. 4*D*), were also greatly up-regulated in E protein-deficient Treg cells. Thus, loss of E protein in differentiated Treg cells affect the expression of a large proportion of genes underlying the multifaceted function of effector Treg cells.

Interestingly, one set of genes not up-regulated by TCR stimulation in E protein-deleted cells are genes associated with IL-2 and IL-7 signaling, i.e., genes that have been shown to be critically involved in initial Treg cell induction and maintenance (23). In particular, neither CD25, CD122 (IL-2Rα and IL-2Rβ chains, respectively), CD127 (IL-7Rα chain) expression was significantly altered upon E-protein depletion in matured Treg cells (*SI Appendix*, Fig. S4*A*) nor was IL-2 signaling as evidenced by phosphorylation of STAT5 (*SI Appendix*, Fig. S3*A*). In addition, c-Rel, a pioneering transcription factor that initiates Treg cell development was also not modulated by E-protein depletion (*SI Appendix*, Fig. S3*B*). It is thus apparent that while decreased E-protein activity is a key feature of the cell signaling pathways



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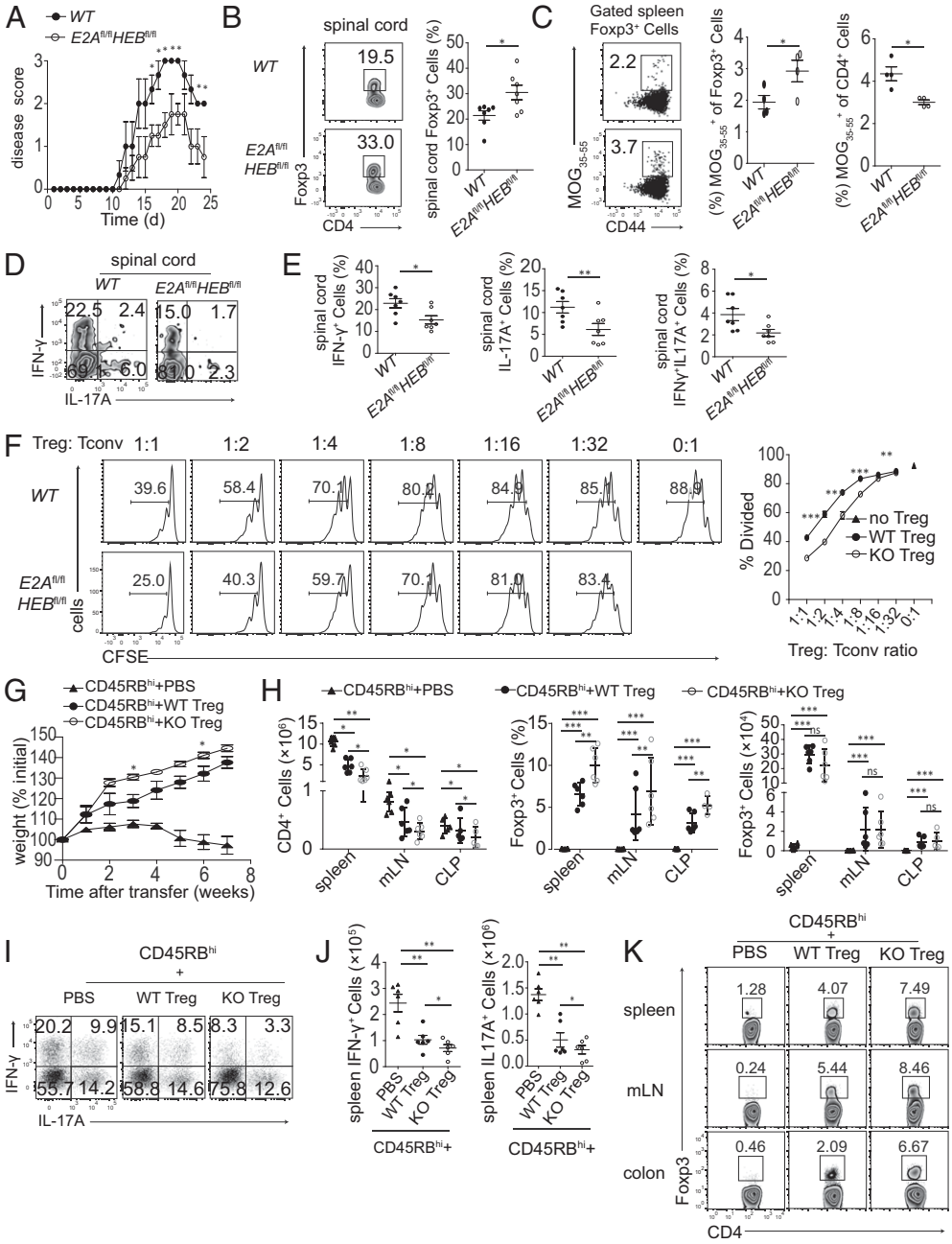


**Fig. 3.** Treg cells deficient in E protein exhibit enhanced immunosuppressive function in vivo. (*A–E*) EAE was induced in WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice and the extent of disease assessed at day 24. (*A*) The disease score of WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice after EAE induction (WT*Foxp3*$^{Cre}$ $n = 4$, *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ $n = 4$). (*B*) Representative flow cytometric analysis (*Left*) and the frequency (*Right*) of Foxp3$^+$ cells among CD4$^+$ cells infiltrating the spinal cord of WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice. (*C*) Flow cytometric analysis of MOG-specific Treg cells in spleens of WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice (*Left*), frequency of MOG$_{35–55}$$^+$ among Foxp3$^+$ cells (*Middle*), and the frequency of MOG$_{35–55}$$^+$ among CD4$^+$ cells (*Right*). (*D* and *E*) Cells from spinal cord of WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice were stimulated with phorbol 12-myristate 13-acetate (PMA) and ionomycin. (*D*) Cytokine production of the cells analyzed by flow cytometry. (*E*) The frequency of IFN-γ$^+$, IL-17A$^+$, and IFN-γ$^+$IL-17A$^+$ cells. (*B–E*) Data are pooled from two independent experiments; each symbol represents data from one mouse, mean ± SD are shown (*$P < 0.05$, **$P < 0.01$). (*F*) CFSE-labeled Tconv cells (CD4$^+$CD25$^-$CD45RB$^{hi}$) from WT mice were cultured with or without titrated numbers of YFP$^+$ Treg cells from WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ or *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice for 3 d after which cell division was analyzed by flow cytometry. Shown are representative cell proliferation dye dilution profiles of Tconv cells of four independent experiments (*Left*) and percentages of these cells, divided at different Tregs. Tconv cell ratios (*Right*); $n = 4$, mean ± SD are shown (**$P < 0.01$, ***$P < 0.001$). (*G–K*) Six- to 8-wk old *RAG2*-deficient mice were adoptively transferred with WT naive CD45RB$^{hi}$ CD4$^+$ T cells together with either PBS (CD45RB$^{hi}$+PBS) or sorted Treg cells derived from WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ mice (CD45RB$^{hi}$+WT Treg) or from *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice (CD45RB$^{hi}$+KO Treg) ($n = 6$ in each group). (*G*) Body weight curve of mice in the various groups. One-way ANOVA with the Dunnett's multiple comparisons test were used to compare the weight loss curve. *$P <$ 0.05. (*H*) The absolute number of CD4$^+$ cells (*Top*), the frequency of Foxp3$^+$ cells in CD4$^+$ cells (*Middle*), and the absolute number of Foxp3$^+$ cells (*Bottom*) in indicated organs. (*I*) Flow cytometry of IFN-γ$^+$ and IL-17A$^+$ cells in CD4$^+$ cells in the spleen. (*J*) The absolute number of IFN-γ$^+$, IL-17A$^+$, and IFN-γ$^+$IL-17A$^+$ cells. (*K*) Flow cytometry analysis of Foxp3$^+$ cells in indicated organs. (*I* and *K*) Data are representative of two independent experiments. (*H* and *J*) Data are pooled from two independent experiments; each symbol represents data from one mouse, mean ± SD are shown. (One-way ANOVA with the Dunnett's multiple comparisons test were used. ns, not significant, *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$.)



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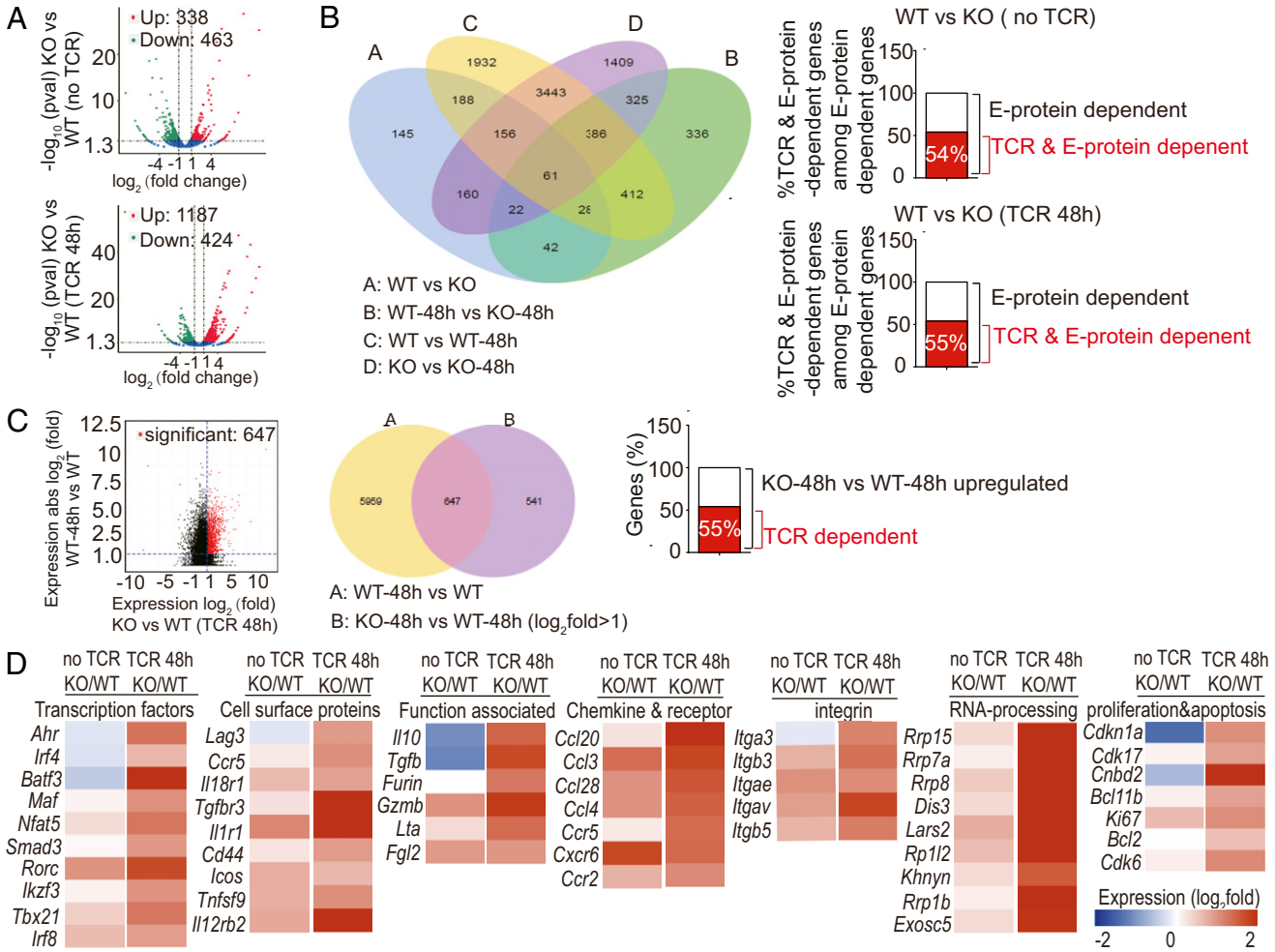


**Fig. 4.** E protein globally regulates effector Treg cell signature gene expression. (*A*) Genes expressed differently in Treg cells from *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice (KO) vs. from WT*Foxp3*$^{Cre}$ mice (WT) (*Top*) or after anti-CD3/CD28 stimulated for 48 h (WT-48 h, KO-48 h, respectively) (*Bottom*) were plotted; numbers in plots indicate genes up-regulated (red) or down-regulated (green) by twofold or more ($P < 0.05$). (*B*, *Left*) Venn diagram. (*Right*) Genes differently expressed twofold or more in KO Treg cells vs. WT Treg cells (Venn A) that also changed expression upon TCR stimulation (Venn C) (*Top*), and genes differently expressed twofold or more in KO-48 h Treg cells vs. WT-48 h Treg cells (Venn B) that were also TCR dependent (Venn C) (*Bottom*). (*C*, *Left*) Genes up-regulated in E-protein KO-48 h cells vs. WT-48 h plotted against those differently expressed in WT-48 h vs. WT cells; (*C*, *Middle*) Venn diagram of genes differently expressed twofold or more in WT-48 h vs. WT Treg cells (TCR-dependent genes) (Venn A) relative to those up-regulated twofold or more in KO-48 h vs. WT-48 h Treg cells (up-regulated genes in E-protein KO Tregs (Venn B); (*C*, *Right*) gene% = TCR-dependent genes among those up-regulated twofold or more in E-protein KO Treg cells. (*D*) Heat map of E protein-regulated genes.

necessary for initial development of Foxp3$^+$ Treg cells, its further decreased expression in matured Treg cells does not regulate these signaling pathways.

In a final set of studies relating to the above RNA-seq studies, we validated the various findings with quantitative RT-PCR analyses. Accordingly, YFP$^+$ Treg cells sorted from spleens of *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ or WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ mice were stimulated with anti-CD3/28 for 48 h after which mRNA extracted from cultured cells was subjected to quantitative RT-PCR. Consistent with the RNA-seq data, a host of molecules essential for effector Treg cell differentiation and function were significantly increased in TCR-stimulated E protein-deficient Treg cells vs. WT Treg cells (*SI Appendix*, Fig. S3*C*). In confirmation, similar studies conducted with Treg cells sorted from *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice vs. cells from WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ mice showed that the expression of most of these genes was decreased in the *Id2*/*Id3*-depleted Treg cells (*SI Appendix*, Fig. S3*D*).

**E-Protein Activity Regulates Effector Treg Cell Differentiation, Trafficking, Proliferation, and Survival.** The observation that the ratio of Foxp3$^+$ cells of *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice increased more remarkably in nonlymphoid tissues than in the spleen or lymph nodes, and that many E protein-dependent genes were associated with effector Treg cell signatures as indicated in RNA-seq analysis, suggested that E protein-deficient Treg cells are prone to differentiate toward effector Treg cells. To confirm the impact of E-protein deletion on effector Treg cell differentiation, we compared *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice and WT *Foxp3*$^{Cre}$ control mice with respect to expression of molecules previously known to associate with Treg cell activation and/or effector function. This analysis revealed that surface molecules indicative of activated or effector Treg cells, such as CD103, ICOS, and KLRG1, were significantly increased in E protein-deficient Treg cells (*SI Appendix*, Fig. S4*A*). As expected, these changes in expression were more pronounced in the CLP and other nonlymphoid organs such as lung and liver, as indicated by the fact that the proportion of Foxp3$^+$ cells that express CD103 and KLRG1 were significantly increased at these tissue sites (Fig. 5 *A* and *B*). Similarly, the expression of ICOS and IRF4, the latter a transcription factor important for effector Treg differentiation and function (24), was also greatly increased



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in Foxp3$^+$ cells from *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice compared with WT mice (Fig. 5*C*).

Treg cells with effector characteristics display an increased ability to accumulate in inflammatory sites to suppress inflammation (25). In a functional study of the effect of E-protein deletion on Treg cell effector characteristics, we determined the effect of deletion of E protein on the ability of Treg cells to accumulate in the spinal cords of mice with EAE, i.e., a site of inflammation where they can exert suppressive function as demonstrated in Fig. 3. We found that in parallel with cells in the spleen (*SI Appendix*, Fig. S4*B*) and draining lymph nodes (*SI Appendix*, Fig. S4*C*) of *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice undergoing EAE, the spinal cords of *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice contained a significantly increased proportion of Treg cells expressing CD103 and KLRG1 (*SI Appendix*, Fig. S4*D*) as well as increased amounts of ICOS and IRF4 (*SI Appendix*, Fig. S4*E*).

In additional studies to determine whether the increased expression of activation/effector markers in Treg cells above were cell intrinsic, we conducted studies of effector Treg cells developing in mixed BM chimeric mice generated as described above and analyzed 8 wk after reconstitution. As expected, we found a higher percentage of CD103$^+$, KLRG-1$^+$ Treg cells (*SI Appendix*, Fig. S4*F*) and up-regulated expression of ICOS and IRF4 in Treg cells (*SI Appendix*, Fig. S4*G*) derived from the *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice donors. In parallel studies, cells from chimeric mice in which EAE had been induced were examined. Here again, the percentage of CD103$^+$Foxp3$^+$ cells or KLRG-1$^+$Foxp3$^+$ cells infiltrating in spinal cord (Fig. 5*D*) or draining lymph nodes (*SI Appendix*, Fig. S4*H*) as well as the expression of ICOS and IRF4 on Treg cells in spinal cord (Fig. 5*E*) or draining lymph nodes (*SI Appendix*, Fig. S4*I*) was greatly increased from the *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice donors than from WT donors. In reciprocal studies to determine if the phenotype of Treg cells from *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ mice was cell intrinsic, we conducted studies of Treg cells in mixed bone marrow chimeras reconstituted with cells from WT *Foxp3*$^{Cre}$ and *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ mice. We found that CD103, ICOS, KLRG1, and IRF4 expression in Treg cells from *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ donor mice were significantly decreased compared with cells from WT *Foxp3*$^{Cre}$ donor mice (Fig. 5 *F* and *G*). These various studies of chimeras thus provided firm evidence that E protein modulates effector Treg cell differentiation in a cell-intrinsic manner.

The RNA-seq analysis also indicated that E protein-deficient Treg cells were also marked by increased expression of genes associated with "apoptosis" and "cell cycle." On the basis of these findings, we reasoned that the increased Treg cell number in peripheral tissue of E protein-deficient mice (Fig. 1) was a consequence of increased Treg cell proliferation or survival in response to continuous autoantigen stimulation. To investigate this possibility, YFP$^+$ Treg cells were sorted from *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ and WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ mice, labeled with CFSE and cultured in the presence of anti-CD3/CD28 and IL-2 for various lengths of time. Treg cells from *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice exhibited a greater number of cell divisions in each time period than WT Foxp3$^+$ cells (Fig. 5*H*) as well as increased numbers of Ki67$^+$ cells (Fig. 5*I*); in addition, they exhibited decreased numbers of AnnexinV$^+$ (apoptotic cells) over 24 h (*SI Appendix*, Fig. S4*J*). Accordingly, this was accompanied by an increased accumulation of Treg cells over the 4-d period of study (Fig. 5*J*).

Collectively, these data provided strong evidence that E protein exerts broad regulation of the gene expression required for effector Treg cell differentiation, function, survival, and proliferation.

**Foxp3$^+$ROR$\gamma$t$^+$ but Not Foxp3$^+$Tbet$^+$ or Foxp3$^+$GATA3$^+$ Effector Treg Cell Subsets Are Regulated by E Protein.** Previous reports indicate that Foxp3$^+$ Treg cells can co-opt the expression of specific transcription factors that are associated with the differentiation and function of effector CD4$^+$ T cell lineages. For example, it has been demonstrated that Foxp3$^+$ T cells expressing ROR$\gamma$t represent a stable regulatory T cell effector lineage with an enhanced suppressive capacity during intestinal inflammation. Similarly, GATA3 and T-bet are expressed in activated Treg cells and are required for Treg cell suppressive function under specific physiological circumstances for the maintenance of Treg cell homeostasis (26–30). Given that ROR$\gamma$t as well as T-bet expression was greatly increased upon TCR stimulation in E protein-deficient Treg cells in the RNA-seq analysis, we investigated whether E protein specifically regulated certain effector Treg cell lineage differentiation. Foxp3$^+$ROR$\gamma$t$^+$, Foxp3$^+$T-bet$^+$, and Foxp3$^+$GATA3$^+$ effector Treg cell subsets were detected in spleen, lung, liver, and CLP of WT *Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice. However, whereas the Foxp3$^+$ROR$\gamma$t$^+$ subset was significantly increased in *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice relative to WT *Foxp3*$^{Cre}$ mice (Fig. 6 *A* and *B*), Foxp3$^+$T-bet$^+$ and Foxp3$^+$GATA3$^+$ Treg cell subsets were not comparably increased (*SI Appendix*, Fig. S5 *A* and *B*). In a series of further studies, we investigated different Treg subset cells under inflammatory conditions, including EAE, DSS-induced colitis, and *Listeria monocytogenes* infection; we reproducibly observed that only Foxp3$^+$ROR$\gamma$t$^+$ Treg cells were dramatically increased in all of these disease conditions (Fig. 6 *C*–*E*), whereas the Foxp3$^+$T-bet$^+$ subset did not change significantly even in the Th1-type *Listeria* infection (Fig. 6*F*). Thus, our findings indicated that E-protein activity specifically regulates Foxp3$^+$ROR$\gamma$t$^+$ Treg cell differentiation. Given that E-protein activity was down-regulated by TCR stimulation, this finding suggests that ROR$\gamma$t$^+$ effector Treg cell differentiation requires relatively stronger TCR stimulation than T-bet$^+$ and GATA3$^+$ Treg cell differentiation.

**E Protein Regulates Effector Treg Cell Signature Gene Expression by Directly Binding to the Regulatory Element of These Genes.** To further investigate the mechanism by which E protein influences gene expression in matured Treg cells, we performed genome-wide ChIP-seq assays. These showed that E2A binds to putative promoter or enhancer regions of a host of genes, including the *Icos*, *Itgae*, *Klrg1*, *Irf4*, *Rorc*, *GranzymeB*, *Il10*, and *Cdk6*, *Bcl2* genes (Fig. 7*A* and *SI Appendix*, Fig. S5*C*). These data were then supported by ChIP-qPCR analysis that showed that E47 binding to genomic fragments containing promoter or conserved enhancer sequences specific for *Itgae*, *Irf4*, *Icos*, *Klrg1*, and *Rorc* (Fig. 7*B*). In related studies we determined patterns of histone methylation of E-protein binding sites in various genes modified by E-protein deletion. We found that trimethylation of histone H3 at Lys4, a histone change consistent with active transcription, was increased in the *Itgae*, *Irf4*, *Icos*, *Klrg1*, and *Rorc* genes of effector Treg cells compared with conventional CD4$^+$ T cells, and the trimethylation of this region was significantly increased in corresponding genes of E protein-deficient Treg cells (Fig. 7*C*). Taken together, these data indicate that E protein directly regulates the expression of genes that are specific for effector Treg cells by directly binding to the regulatory element of these genes and by remodeling of the chromatin at regulatory elements of key genes affecting effector Treg cell differentiation so as to render these sites less transcriptionally active; in contrast, reduced E-protein activity causes these sites to be more transcriptionally active and gives rise to differentiation of effector Treg cells with enhanced suppressor function.

## Discussion

Despite major progress in our understanding of TCR engagement in the differentiation of effector Treg cells, the downstream molecular mechanisms underlying TCR regulation of Treg cell homeostasis and suppressive capacity is still poorly defined. Here we demonstrate that specific deletion of E protein in matured Treg cells (in *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*-GFP-hCre mice) resulted in Treg cells with increased stability/survival and enhanced suppressive function. We also showed that a large proportion of genes regulated by E protein was also TCR dependent, strongly suggesting that the effects of E protein on gene regulation are parallel to or slightly amplified by TCR stimulation. Finally, we showed with ChiP-seq analysis that E protein binds to genes



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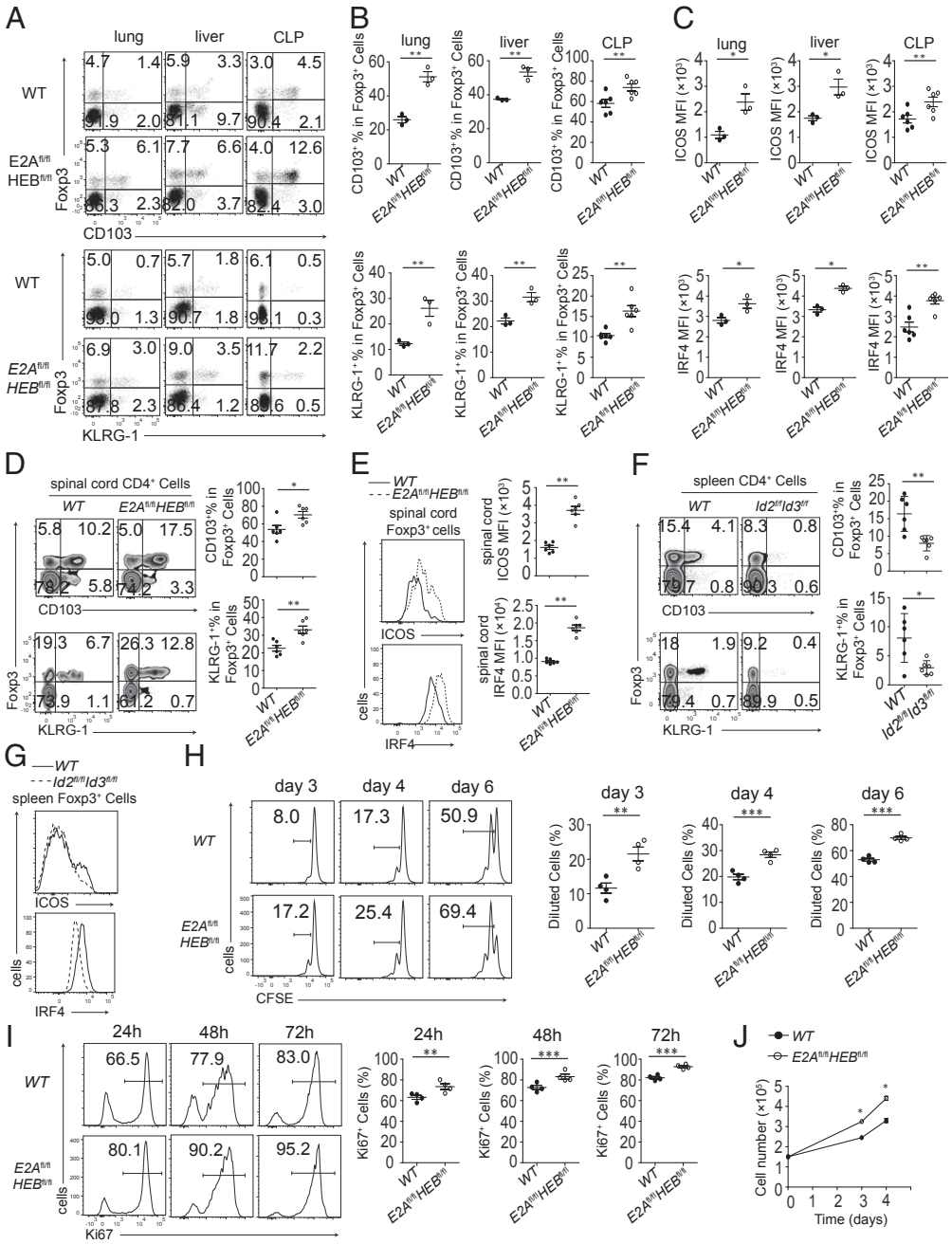


**Fig. 5.** E-protein activity regulates effector Treg cell differentiation, trafficking, proliferation, and survival. (*A*) Cells of indicated organs from WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice were stained directly with CD4, Foxp3, CD103, and KLRG-1. Dot plots were gated on CD4$^{+}$ cells. (*B*) Frequencies of CD103$^{+}$ (*Top*) and KLRG-1$^{+}$ (*Bottom*) cells in Foxp3$^{+}$ cells in indicated organs from WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice. (*C*) Mean fluorescence of ICOS (*Top*) and IRF4 (*Bottom*) on Foxp3$^{+}$ cells in indicated organs of WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice. (*D* and *E*) Mixed BM chimeric mice were generated by transferring BM cells from CD45.1$^{+}$CD45.2$^{+}$ WT *Foxp3*$^{Cre}$ and CD45.1$^{+}$ *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice into lethally irradiated recipient mice. (*D*) Flow cytometric analysis of CD103, KLRG-1, and Foxp3 expression among CD4$^{+}$ cells (*Left*) and the frequencies of CD103$^{+}$ (*Right Top*) and KLRG-1$^{+}$ (*Right Bottom*) cells among CD45.1$^{+}$CD45.2$^{+}$ WT *Foxp3*$^{Cre}$ and CD45.1$^{+}$ *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ Foxp3$^{+}$ cells in the spinal cord of BM chimeric mice 24 d after EAE induction. (*E*) Flow cytometric analysis of ICOS and IRF4 expression (*Left*) and the mean fluorescence of ICOS (*Right Top*) and IRF4 (*Right Bottom*) among CD45.1$^{+}$CD45.2$^{+}$ WT *Foxp3*$^{Cre}$ and CD45.1$^{+}$ *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ Treg cells in the spinal cord of BM chimeric mice 24 d after EAE induction. (*F* and *G*) Mixed BM chimeric mice were generated by transferring BM cells from WT *Foxp3*$^{Cre}$ and *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ mice into lethally irradiated recipient mice. (*F*) Flow cytometric analysis of CD103, KLRG-1, and Foxp3 expression among CD4$^{+}$ cells (*Left*) and the frequencies of CD103$^{+}$ (*Right Top*) and KLRG-1$^{+}$ (*Right Bottom*) cells among WT *Foxp3*$^{Cre}$ and *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ Foxp3$^{+}$ cells in the spleen of the mixed BM chimeric mice. (*G*) Flow cytometric analysis of ICOS and IRF4 expression of WT *Foxp3*$^{Cre}$ and *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$*Foxp3*$^{Cre}$ Foxp3$^{+}$ cells in the spleen of the mixed BM chimeric mice. (*H–J*) Highly purified Treg cells from WT *Foxp3*$^{Cre}$*R26*$^{YFP}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$*R26*$^{YFP}$ mice were (*H*) labeled with CFSE and stimulated with anti-CD3 plus anti-CD28 in the presence of human IL-2 for the indicated time. CFSE dilution in the Treg cells at the indicated time points were analyzed by flow cytometry (*Left*) and the frequencies of CFSE-diluted cells (*Right*); or (*I*) Ki67$^{+}$ in Foxp3$^{+}$ cells at the indicated time (*Left*) and the frequencies of Ki67$^{+}$ in Foxp3$^{+}$ cells (*Right*). (*J*) Total cell numbers at the indicated time points were counted separately. (*A*, *G* and *D*, *E*, *F*, *H*, and *I*, *Left*) Data are representative of at least three independent experiments. (*B*, *C*, *J* and *D*, *E*, *F*, *H*, and *I*, *Right*) Data are pooled from at least three independent experiments, each symbol represents data from one mouse, mean ± SD are shown (*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$).



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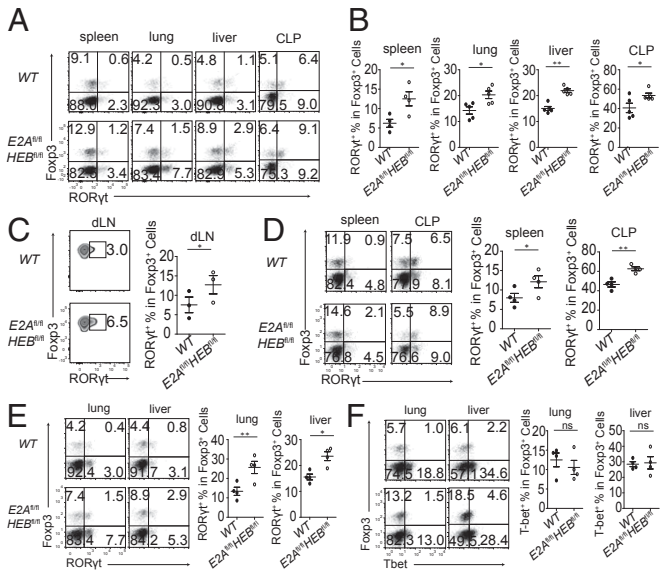


**Fig. 6.** Foxp3$^+$RORγt$^+$ but not Foxp3$^+$Tbet$^+$ or Foxp3$^+$GATA3$^+$ effector Treg cell subset is regulated by E protein. (*A* and *B*) Flow cytometric analysis of RORγt and Foxp3 expression in cells from indicated organs from WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice (*A*) and the frequencies of RORγt$^+$ cells within Treg cells (*B*). (*C*) Flow cytometric analysis of RORγt expression in Foxp3$^+$ cells in the draining lymph node from WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice 21 d after EAE induction (*Left*) and the frequency of RORγt$^+$ cells within these Treg cells (*Right*). (*D*) DSS-colitis was induced in WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice; flow cytometric analysis of RORγt and Foxp3 expression in indicated organs (*Left*) and the frequencies of RORγt$^+$ cells in Treg cells 10 d after colitis induction (*Right*). (*E* and *F*) WT*Foxp3*$^{Cre}$ and *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*$^{Cre}$ mice were infected by *L. monocytogenes*; (*E*) flow cytometric analysis of RORγt and Foxp3 expression and the frequencies of RORγt$^+$ cells in isolated lung and liver Treg cells (*Right*). (*F*) Flow cytometric analysis of T-bet and Foxp3 expression (*Left*) and the frequencies of T-bet$^+$ cells among Treg cells (*Right*). (*A* and *C–F*, *Left*) Data are representative of at least three independent experiments. (*B* and *C–F*, *Right*) Data are pooled from at least three independent experiments; each symbol represents data from one mouse, mean ± SD were shown (ns, not significant, $*P < 0.05$, $**P < 0.01$).

involved in Treg cell effector function and that decreased binding (resulting from down-regulation) leads to enhanced expression of these genes. Thus, TCR signaling regulates Treg cell function by releasing these genes from the negative effects of E protein.

Regarding the specific effects of E proteins, the above-mentioned ChIP-seq studies showed that before down-regulation by TCR signaling, E proteins bind to a large number of the genes that are associated with effector Treg cell differentiation (*Irf4*, *Rorc*), localization (*Itgae* and *Ccr5*, *Ccr2*, etc.), function (*Icos*, *Lag3*, *Il10*, and *GranzymeB*), survival (*Bcl2*), and proliferation (*Cdk6*); in addition, E proteins bind to the gene encoding *Klrg1*, a marker of mature or terminally differentiated cells, including Treg cells. In studies probing the effects of E-protein binding we found that histone changes indicative of active gene transcription (trimethylation at Lys4 of H3) were present in many of the above genes following E-protein down-regulation. Thus, the picture that emerges is that E proteins have a potent capacity to suppress transcription of those genes necessary for effector Treg cell function. Of particular interest in this context is that IRF4 is dramatically up-regulated during inflammation in E protein-deficient Treg cells, indicating that E protein negatively regulates the expression IRF4. This observation is significant because IRF4 is a TCR-induced gene that has been shown to be critical for the optimal function of effector Treg cells and to be a mediator of effector Treg cell differentiation; thus, this finding greatly strengthens the link between Treg cell differentiation and TCR-regulated E-protein activity.

E-protein regulation of Treg cells is pertinent to the generation of RORγt$^+$Foxp3$^+$ cells, a functionally distinct Treg cell subset that has been shown to suppress Th17 responses in EAE (22). In the present study we showed that RORγt$^+$ Foxp3$^+$ Treg cells were significantly increased in mice with E protein-deficient Treg cells, both in the steady state and in inflammatory states. In addition, we found that the gene expression patterns of E protein-deficient Treg cells are very similar to that of RORγt$^+$ Foxp3$^+$ cells found in the intestine and in immunization-induced RORγt$^+$ Foxp3$^+$cells (22) since in all cases one sees up-regulation of the *Icos*, *Itgae*, *Rorc*, *Il1r1*, *Il23r*, *Havrc2*, and *Il10* genes. Finally, E protein-deficient Treg cells exhibited enhanced suppressive activity and stability, also in line with characteristics of previously described RORγt$^+$ Treg cells. These findings suggest that E protein is a key regulator of the RORγt$^+$ Treg cell subset and that this subset arises when Tregs are subjected to a sufficient and possibly higher level of TCR signaling and E-protein deletion that allows transcription of the *RORγt* gene.

Recent studies of mice with Id2/Id3 deficiency in Foxp3$^+$ Treg cells provide data in some ways parallel to those obtained in this study in that these mice also manifest E-protein abnormalities (16). However, in this case increased E-protein function results in decreased Treg cell stability and in vivo suppressive function, the latter manifesting as spontaneous inflammation of the lungs, eyelids, skin, and esophagus (16). These changes at first glance suggest that Id2/Id3 deletion provides a mirror image of E-protein deletions and thereby predicts the effects of E-protein deletion. However, this is probably not the case for several reasons: first, Id2/Id3 deletion impacts mainly on regulation of B cell/Th2 responses and results in allergic manifestations; in contrast, E-protein deletion has a more general effect on Treg cell suppressive function and leads to enhanced suppression of Th1/Th17-mediated inflammation in EAE and transfer colitis;

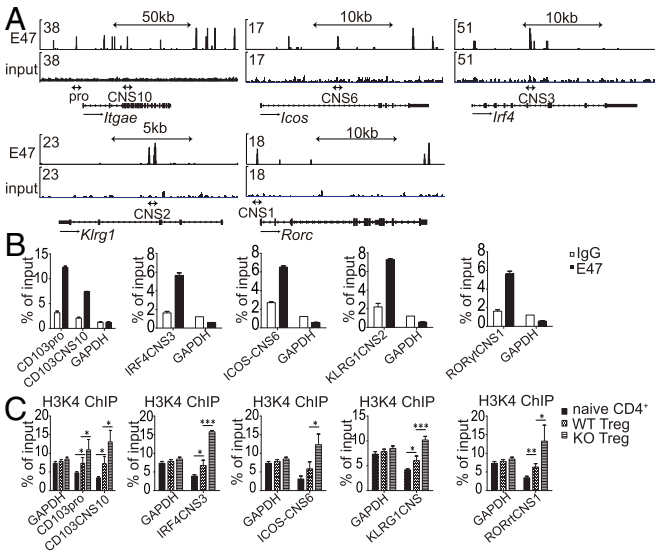


**Fig. 7.** E protein regulates effector Treg cell signature gene expression by directly binding to the regulatory element of these genes. (*A*) Thymocytes were isolated from WT mice; E47 ChIP was performed and subjected to library preparation and sequencing as described in *Methods*. Representative of ChIP sequencing tracks showing composite binding of E2A to *Itgae*, *Icos*, *Irf4*, *Klrg1*, and *Rorc* genes. (*B*) Chromatin from thymocytes were subjected to immunoprecipitation with anti-E47. Binding of E proteins to the indicated genes was analyzed by quantitative real-time PCR. Results are representative of at least three independent experiments; mean ± SD are shown. (*C*) Chromatin from purified naive CD4$^+$ cells, WT Treg cells, and E protein-deficient Treg cells were subjected to immunoprecipitation with anti-H3K4. Binding of H3K4 to the indicated genes was analyzed by quantitative real-time PCR. Results are representative of two independent experiments; mean ± SD are shown ($*P < 0.05$, $**P < 0.01$, $***P < 0.001$).

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second, because deletion of Id2/Id3 in matured Treg cells leads to spontaneous autoimmune disease, the apparent effect of Id2/Id3 on Treg cell function derived from deletion studies may in reality be secondary to inflammatory effects of such deletion; third, TCR stimulation of Id protein-deleted cells results in both down-regulation of Id and E-protein expression rather than a reciprocal change that one might expect in the case of opposing functions. These various differences between the effects of Id2/Id3 and E-protein deletion are likely to arise from the fact that at the molecular level there does not exist a strict point/counterpoint relation between these sets of transcriptional regulatory factors. For this reason, it is fair to say that direct deletion of E protein provides a better picture of E-protein effects on Treg cells than the indirect effects of Id protein deletion.

Prior studies have revealed that E-protein expression regulates cell proliferation and survival (apoptosis) (31). This correlates with the results of RNA-seq analyses performed in this study that showed that genes involved in the "cell proliferation and cycling" such as *Bcl2*, *Cdk6*, and *Cdk14* genes, were up-regulated in E protein-deficient Treg cells. In addition, the latter cells exhibited increased proliferative responses in vitro upon TCR stimulation, which most likely contributed to the increased percentage of effector Treg cells noted in *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$*Foxp3*-GFP-hCre mice in vivo. These findings correlated with the results of ChIP-seq analysis that revealed that genes involved in cell cycling exhibited a high level of E-protein binding, including the *Bcl2* gene and cycle promoter gene, *Cdk6*; it is thus likely that the increased proliferation and up-regulated expression of cell cycle genes were also a major functional consequence of E-protein loss in Treg cells.

In summary, these studies provide strong evidence that E protein regulates a large set of key effector Treg cell signature genes that govern the differentiation, effector function, proliferation, and survival of already established Treg cells. In as much as E-protein down-regulation is largely under TCR control, these findings indicate that the molecular mechanism of TCR regulation of Treg cell differentiation is in fact its ability to control E-protein activity. Thus, E protein plays a stage-dependent role in Treg cells, whereby it is not only indispensable for their development in the thymus as demonstrated previously (15), but also essential for their further differentiation into effector Treg cells in the periphery. On this basis, the underlying regulatory circuits governing E-protein levels are likely to provide insight into how Treg cells maintain their essential identity and at the same time exhibit plasticity in various tissue settings.

## Methods

**Mice.** WT CD45.1 mice were obtained from The Jackson Laboratory. *Tcf3*(E2A)$^{fl/fl}$*Tcf12*(HEB)$^{fl/fl}$ mice, *Id2*$^{fl/fl}$*Id3*$^{fl/fl}$ mice, and *Foxp3*-GFPKI mice were gifts from Y. Zhuang, Duke University, Durham, NC, and from M. Oukka, Brigham and Women's Hospital, Boston, MA, respectively. *Foxp3*$^{Cre}$/*Rosa26*$^{YFP}$ mice were a gift from Xuyu Zhou, Chinese Academy of Sciences, Beijing, China. Strain-matched WT *Foxp3*$^{Cre}$ mice served as controls. *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$ mice were crossed with *Foxp3*-GFP-KI mice and *Foxp3*$^{Cre}$/*Rosa26*$^{YFP}$ mice to generate *E2A*$^{fl/fl}$*HEB*$^{fl/fl}$ Foxp3-GFP-KI/*Foxp3*$^{Cre}$/*Rosa26*$^{YFP}$ mice. The study was approved by the Research Ethics Committee of the Institute of Microbiology, Chinese Academy of Sciences (IMCAS), the permit number was APIMCAS2017015.

**Flow Cytometry and Cell Sorting.** Flow cytometry was performed on a FACSCalibur or FACS CantoII (BD Biosciences) and the data were analyzed by FlowJo. Cell sorting was performed using a FACS AriaIII (BD). Fluorochrome-conjugated Abs are provided in *SI Appendix*.

An extended methods section is provided in *SI Appendix, SI Materials and Methods*.

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1. Sakaguchi S, Yamaguchi T, Nomura T, Ono M (2008) Regulatory T cells and immune tolerance. *Cell* 133:775–787.
2. Curotto de Lafaille MA, Lafaille JJ (2009) Natural and adaptive foxp3+ regulatory T cells: More of the same or a division of labor? *Immunity* 30:626–635.
3. Campbell DJ, Koch MA (2011) Phenotypical and functional specialization of FOXP3+ regulatory T cells. *Nat Rev Immunol* 11:119–130.
4. Lee JH, Kang SG, Kim CH (2007) FoxP3+ T cells undergo conventional first switch to lymphoid tissue homing receptors in thymus but accelerated second switch to non-lymphoid tissue homing receptors in secondary lymphoid tissues. *J Immunol* 178:301–311.
5. Panduro M, Benoist C, Mathis D (2016) Tissue Tregs. *Annu Rev Immunol* 34:609–633.
6. Kleinewietfeld M, et al. (2005) CCR6 expression defines regulatory effector/memory-like cells within the CD25(+)CD4+ T-cell subset. *Blood* 105:2877–2886.
7. Huehn J, et al. (2004) Developmental stage, phenotype, and migration distinguish naive- and effector/memory-like CD4+ regulatory T cells. *J Exp Med* 199:303–313.
8. Suffia I, Reckling SK, Salay G, Belkaid Y (2005) A role for CD103 in the retention of CD4+CD25+ Treg and control of Leishmania major infection. *J Immunol* 174:5444–5455.
9. Beyersdorf N, Ding X, Tietze JK, Hanke T (2007) Characterization of mouse CD4 T cell subsets defined by expression of KLRG1. *Eur J Immunol* 37:3445–3454.
10. Lee HM, Bautista JL, Scott-Browne J, Mohan JF, Hsieh CS (2012) A broad range of self-reactivity drives thymic regulatory T cell selection to limit responses to self. *Immunity* 37:475–486.
11. Vahl JC, et al. (2014) Continuous T cell receptor signals maintain a functional regulatory T cell pool. *Immunity* 41:722–736.
12. Levine AG, Arvey A, Jin W, Rudensky AY (2014) Continuous requirement for the TCR in regulatory T cell function. *Nat Immunol* 15:1070–1078.
13. Murre C, McCaw PS, Baltimore D (1989) A new DNA binding and dimerization motif in immunoglobulin enhancer binding, daughterless, MyoD, and myc proteins. *Cell* 56:777–783.
14. Maruyama T, et al. (2011) Control of the differentiation of regulatory T cells and T(H)17 cells by the DNA-binding inhibitor Id3. *Nat Immunol* 12:86–95.
15. Gao P, et al. (2014) Dynamic changes in E-protein activity regulate T reg cell development. *J Exp Med* 211:2651–2668.
16. Miyazaki M, et al. (2014) Id2 and Id3 maintain the regulatory T cell pool to suppress inflammatory disease. *Nat Immunol* 15:767–776.
17. Zhou X, et al. (2008) Selective miRNA disruption in T reg cells leads to uncontrolled autoimmunity. *J Exp Med* 205:1983–1991.
18. Zhou X, et al. (2009) Instability of the transcription factor Foxp3 leads to the generation of pathogenic memory T cells in vivo. *Nat Immunol* 10:1000–1007.
19. Thornton AM, et al. (2010) Expression of Helios, an Ikaros transcription factor family member, differentiates thymic-derived from peripherally induced Foxp3+ T regulatory cells. *J Immunol* 184:3433–3441.
20. Sefik E, et al. (2015) MUCOSAL IMMUNOLOGY. Individual intestinal symbionts induce a distinct population of RORγ$^+$ regulatory T cells. *Science* 349:993–997.
21. Yang BH, et al. (2016) Foxp3(+) T cells expressing RORγt represent a stable regulatory T-cell effector lineage with enhanced suppressive capacity during intestinal inflammation. *Mucosal Immunol* 9:444–457.
22. Kim BS, et al. (2017) Generation of RORγt$^+$ antigen-specific T regulatory 17 cells from Foxp3$^+$ precursors in autoimmunity. *Cell Rep* 21:195–207.
23. Setoguchi R, Hori S, Takahashi T, Sakaguchi S (2005) Homeostatic maintenance of natural Foxp3(+) CD25(+) CD4(+) regulatory T cells by interleukin (IL)-2 and induction of autoimmune disease by IL-2 neutralization. *J Exp Med* 201:723–735.
24. Zheng Y, et al. (2009) Regulatory T-cell suppressor program co-opts transcription factor IRF4 to control T(H)2 responses. *Nature* 458:351–356.
25. Liston A, Gray DH (2014) Homeostatic control of regulatory T cell diversity. *Nat Rev Immunol* 14:154–165.
26. Wohlfert EA, et al. (2011) GATA3 controls Foxp3$^+$ regulatory T cell fate during inflammation in mice. *J Clin Invest* 121:4503–4515.
27. Wang Y, Su MA, Wan YY (2011) An essential role of the transcription factor GATA-3 for the function of regulatory T cells. *Immunity* 35:337–348.
28. Yu F, Sharma S, Edwards J, Feigenbaum L, Zhu J (2015) Dynamic expression of transcription factors T-bet and GATA-3 by regulatory T cells maintains immunotolerance. *Nat Immunol* 16:197–206.
29. Koch MA, et al. (2009) The transcription factor T-bet controls regulatory T cell homeostasis and function during type 1 inflammation. *Nat Immunol* 10:595–602.
30. Levine AG, et al. (2017) Stability and function of regulatory T cells expressing the transcription factor T-bet. *Nature* 546:421–425.
31. Bain G, Quong MW, Soloff RS, Hedrick SM, Murre C (1999) Thymocyte maturation is regulated by the activity of the helix-loop-helix protein, E47. *J Exp Med* 190:1605–1616.



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